
An Integrated Clinical–Imaging Reporting Workflow for the Assessment of Cognitive Impairment

Abstract

The quantitative assessment of brain MRI biomarkers is crucial for diagnosing Alzheimer’s disease (AD) and Mild Cognitive Impairment (MCI) but remains a bottleneck in clinical workflows. While deep learning models have shown promise, they often lack the interpretability and structured reporting capabilities required for clinical adoption. In this study, we propose an integrated, automated clinical–imaging reporting workflow that combines deep learning-based segmentation (FastSurfer) with normative modeling and LLM(Large Language Models). Our pipeline extracts regional volumes, computes Z-scores, and utilizes GPT-5.2 to generate structured radiology reports. We validated this workflow on 2,576 subjects across three multi-center cohorts (OASIS-3, NACC, AJU). The system successfully processed 83.1% of scans and demonstrated high scalability, generating 3,963 reports at a rate of 8.6 reports per minute. Crucially, a systematic audit of 200 subjects (1,200 regional assessments) revealed a quantitative-to-text concordance rate of 83.7%. The discrepancies were exclusively characterized by "conservative bias" (underreporting severity), with zero instances of overreporting or hallucination. These results demonstrate that our framework operates with a "safe failure mode," providing transparent, efficient, and clinically safe decision support.

1 Introduction

Quantitative analysis of structural brain MRI plays a crucial role in the clinical assessment of neurodegenerative and cognitive disorders. Volumetric measurements of regions such as the hippocampus and entorhinal cortex have been established as sensitive biomarkers for Alzheimer’s disease (AD) and Mild Cognitive Impairment (MCI) [1]. However, the process of efficiently translating imaging biomarkers into routine clinical reports remains limited. Radiologists often rely on visual inspection or manual measurements, which are time-consuming and susceptible to inter-observer variability [2].

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The necessity of quantitative reporting in dementia imaging has been well established. Pemberton et al. demonstrated that providing automated volumetric reports significantly improved radiologists’ diagnostic sensitivity (from 66% to 75%) and confidence compared to visual assessment alone [3]. However, while such systems successfully deliver quantitative evidence in tabular or graphical formats, the final step—converting these metrics into structured, narrative radiology reports—has remained a manual task requiring cognitive synthesis by the reading physician. This documentation bottleneck limits the scalability of quantitative imaging in high-throughput clinical workflows.

A primary impediment to automating clinical reporting is the “black box” nature of end-to-end deep learning models. As Rudin argues, high-stakes domains like healthcare require inherent interpretability and auditability rather than post-hoc explanations [4]. While LLMs have demonstrated remarkable capabilities in text generation, their application to quantitative medical imaging raises concerns about grounding generated text in precise numerical evidence and preventing fabrication of non-existent findings (hallucinations).

To address these challenges, we propose an automated, rule-based clinical–imaging reporting workflow that extends quantitative volumetry to structured narrative generation. Instead of training a black-box diagnostic classifier, our approach integrates established segmentation tools (FastSurfer) with statistical normative modeling and constraint-based prompting of GPT-5.2. By feeding age- and sex-adjusted Z-scores as structured input to the LLM, we aim to generate radiology-style reports that are explicitly grounded in verifiable quantitative evidence, thereby combining the efficiency of automation with the transparency required for clinical deployment.

We validate this framework using AD and MCI as representative clinical applications, where structural brain atrophy and interpretation of age-adjusted deviations are clinically central. Across three multi-center cohorts (OASIS-3, NACC, AJU), we evaluate the feasibility, reliability, and auditability of the proposed workflow under diverse imaging environments.

Our contributions are as follows:

- We present a pipeline that integrates segmentation, normative scoring, and language generation to facilitate the transition from raw MRI to clinically interpretable, structured, and auditable radiology reports grounded in quantitative evidence.
- We demonstrate the construction and application of an age- and sex-adjusted normative database to enhance the interpretability and transparency of volumetric measurements.
- We validate the workflow’s feasibility across multi-center datasets, demonstrating that it can produce detailed, evidence-based reports in under 3 seconds per patient, while maintaining a conservative error profile that mitigates fabrication risks.

Alzheimer’s disease (AD) presents a rapidly escalating challenge to global public health, with prevalence estimates projected to triple by 2050 [1]. The current clinical standard for diagnosing AD and its prodromal stage, Mild Cognitive Impairment (MCI), relies heavily on the identification of structural atrophy in brain MRI. However, the quantitative assessment of these biomarkers creates a significant bottleneck. Radiologists must often rely on visual inspection or manual measurements, processes that are time-consuming and prone to inter-observer variability [2].

A primary impediment to the clinical deployment of AI is the “black box” nature of end-to-end deep learning models. As Rudin argues, high-stakes domains like healthcare require inherent interpretability rather than post-hoc explanations [4]. While LLMs have demonstrated remarkable capabilities in text generation, their application to volumetric 3D medical data remains challenging due to the difficulty of grounding generated text in precise quantitative metrics.

To address these challenges, we propose an automated, rule-based clinical–imaging reporting workflow. Instead of training a black-box diagnostic classifier, our approach integrates established segmentation tools with statistical normative modeling and the generative capabilities of GPT-5.2. We utilize FastSurfer for rapid, consistent parcellation and compare derived volumes against a normative database to calculate Z-scores. These quantitative metrics serve as structured prompts for GPT-5.2, with the aim of grounding the generated radiology reports in volumetric evidence.

Our contributions are as follows:

- We present a pipeline that integrates segmentation, normative scoring, and language generation to facilitate the transition from raw MRI to structured clinical reports.

- We demonstrate the construction of a normative database to provide age- and sex-adjusted Z-scores, enhancing the interpretability of brain atrophy measurements.
- We validate the workflow’s feasibility across multi-center datasets, showing that it can produce detailed, evidence-based reports in under 3 seconds per patient, offering a practical solution for scalable clinical analysis.

2 Related Work

2.1 Multimodal Deep Learning for Alzheimer’s Diagnosis

Deep learning has established itself as the standard for analyzing neuroimaging data in Alzheimer’s disease (AD) diagnosis. Early approaches primarily utilized 3D Convolutional Neural Networks (CNNs) to extract features from structural MRI for binary classification (e.g., AD vs. Normal) [2, 5]. To enhance diagnostic precision, recent studies have shifted towards multimodal fusion frameworks. For instance, Qiu et al. demonstrated that integrating MRI data with clinical demographics and genetic markers (e.g., APOE) significantly outperforms unimodal baselines [6]. However, these fully supervised methods rely heavily on large-scale annotated datasets. In clinical reality, acquiring thousands of high-quality, labeled 3D MRI scans with corresponding genetic data is often unfeasible. Furthermore, these models typically function as "black boxes," outputting probability scores without the quantitative volumetric measurements (e.g., hippocampal volume in mm^3) required for radiological reporting.

2.2 Limits of Current Medical VLMs

The advent of Foundation Models has catalyzed the development of Medical VLMs capable of processing diverse modalities [7]. Adapted from general-domain models like LLaVA [8], systems such as MedCLIP and the recent MIMO [9] have demonstrated capabilities in visual question answering and report generation. While MIMO introduced pixel-level grounding to link text with image regions, these models are largely optimized for 2D imaging (e.g., Chest X-rays, Pathology slides). They struggle to encode the spatial complexity of 3D volumetric data and, more critically, fail to perform precise regression tasks needed to quantify brain atrophy. Our work bridges this gap by proposing a **quantitative-metric-guided LLM reporting system** that utilizes normative data to generate quantitative volumetric evidence alongside natural language reports.

2.3 Clinical Utility of Quantitative Reporting and Generative AI

The integration of AI into clinical workflows relies heavily on two pillars: interpretability and efficiency. Pemberton et al. (2021) demonstrated that providing automated quantitative volumetry reports (QReports) significantly improves radiologists’ diagnostic sensitivity (from 66% to 75%) and confidence compared to visual inspection alone [3]. While their study confirmed the value of objective volumetric evidence in reducing inter-reader variability, the output remained limited to visual plots and tabular data.

Parallel to this, Nakaura et al. (2024) explored the potential of LLMs in documentation, showing that GPT-4 could generate radiology reports from concise findings with accuracy comparable to board-certified radiologists [10]. However, their work primarily focused on qualitative summarization of CT findings. Our research synthesizes these two paradigms: we utilize the rigorous quantitative evidence advocated by Pemberton et al. as the structured input for an LLM, thereby leveraging the generative capabilities validated by Nakaura et al. to produce comprehensive, evidence-based narrative reports for 3D MRI analysis.

3 Methods

3.1 Study Population and Datasets

We analyzed T1-weighted brain MRI scans from 2,576 subjects across three independent cohorts (Table I). The **OASIS-3** dataset [11] contributed 448 subjects from Washington University in St. Louis, with scans provided in skull-stripped NIfTI format following the Brain Imaging Data Structure

(BIDS) standard. The **NACC** cohort [12] included 1,122 subjects from multiple Alzheimer’s Disease Research Centers in the United States, with raw DICOM images covering the full head and neck. The **AJU** cohort comprised 1,006 subjects from Ajou University Hospital, South Korea, with a high proportion of MCI cases, providing data enriched for early-stage disease.

Diagnostic classifications followed standardized criteria for CN, MCI, and AD subjects [13]. The final dataset showed a balanced distribution across disease stages (36% CN, 44% MCI, 20% AD).

Table 1: Characteristics of the multi-center neuroimaging datasets. CN: cognitively normal, MCI: mild cognitive impairment, AD: Alzheimer’s disease.

Characteristic	OASIS-3	NACC	AJU	Total
<i>Sample Size</i>				
Total subjects	448	1,122	1,006	2,576
Successfully processed	414 (92.4%)	834 (74.3%)	892 (88.7%)	2,140 (83.1%)
<i>Diagnostic Distribution</i>				
CN	363 (81%)	539 (48%)	30 (3%)	932 (36%)
MCI	49 (11%)	325 (29%)	765 (76%)	1,139 (44%)
AD	36 (8%)	258 (23%)	211 (21%)	505 (20%)
<i>Data Format</i>				
Image format	NIfTI (BIDS)	DICOM	DICOM	—
Skull-stripped	Yes	No	No	—
Neck included	No	Yes	Yes	—
<i>MRI Acquisition</i>				
Sequence	3D T1w MPRAGE	3D T1w	3D T1w	—
Field strength	3T	1.5T/3T	3T	—
Resolution (mm ³)	1.0 isotropic	Variable	1.0 isotropic	—

3.2 Preprocessing Pipeline

All images were processed using a standardized multi-center preprocessing pipeline. DICOM images from NACC and AJU were converted to NIfTI format, and full-head acquisitions underwent neck cropping to remove inferior non-brain structures. Brain extraction was performed using FSL BET, followed by automated cortical and subcortical parcellation with FastSurfer to obtain volumetric labels. Finally, all images and label maps were registered to the MNI152 1 mm template using affine normalization, ensuring spatial comparability of regional volumes across cohorts.

3.3 Volumetric Analysis and Normative Modeling

Regional brain volumes were extracted from FastSurfer-derived parcellation maps. To ensure clinical relevance, we specifically focused on **95 key anatomical structures** known to be biomarkers for Alzheimer’s disease.

These included the **medial temporal lobe** (hippocampus, amygdala, entorhinal cortex) for memory assessment, **subcortical structures** (thalamus, caudate, putamen), and the **ventricular system** (lateral and inferior lateral ventricles) to quantify global and local atrophy rates.

Volumes were computed in both native and MNI-normalized space to enable consistency checks and downstream group-level analyses. A normative reference database was constructed from cognitively normal (CN) subjects, stratified by age (5-year bins) and sex. This database was used to derive standardized Z-scores and percentile ranks for each region, forming the quantitative basis for the subsequent automated reporting module.

To translate continuous volumetric data into clinically actionable insights, age- and sex-adjusted Z-scores were mapped to discrete atrophy grades. We adopted percentile-based thresholds aligned with FDA-cleared volumetric software standards (Table 2).

Table 2: Standardized atrophy grading thresholds derived from FDA-cleared reference standards (e.g., NeuroQuant). Ventricular enlargement uses an inverse scale.

Grade	Percentile Range	Z-score Range	Clinical Interpretation
Normal	>16th	> -1.0	Within normal limits
Mild atrophy	10th–16th	-1.0 to -1.28	Borderline, monitor progression
Moderate atrophy	5th–10th	-1.28 to -1.65	Clinically significant
Severe atrophy	≤5th	≤ -1.65	Marked volume loss
<i>Ventricular Enlargement (Inverse Scale)</i>			
Normal	<84th	< +1.0	Within normal limits
Mild enlargement	84th–90th	+1.0 to +1.28	Borderline
Moderate enlargement	90th–95th	+1.28 to +1.65	Ex vacuo dilation likely
Severe enlargement	≥95th	≥ +1.65	Marked enlargement
Reference: FDA 510(k) K170981; Percentile thresholds adapted from Kovacevic et al. (2009) [14]			

Specifically for the medial temporal lobe, hippocampal percentiles were converted into an automated Medial Temporal Atrophy (MTA) score (Table 3). This scoring system incorporates age-dependent cutoffs to distinguish pathological atrophy from normal aging. Furthermore, bilateral asymmetry indices and hippocampus–ventricle ratios were computed to capture complex neurodegenerative patterns, such as asymmetric frontotemporal atrophy or AD-typical profiles.

Table 3: Automated Medial Temporal Atrophy (MTA) scoring logic based on hippocampal percentiles and age-adjusted interpretation rules.

MTA Score	Hippocampal Percentile	Description
0	>25th	No atrophy
1	10th–25th	Mild widening of choroid fissure
2	5th–10th	Widening of choroid fissure + mild temporal horn dilation
3	2nd–5th	Moderate hippocampal volume loss + temporal horn dilation
4	≤2nd	Severe hippocampal atrophy
<i>Age-Adjusted Normal Cutoffs</i>		
<65 years	—	MTA ≤1 considered normal
65–74 years	—	MTA ≤1.5 considered normal
≥75 years	—	MTA ≤2 considered normal
Reference: Scheltens et al. (1992) [15]; Age cutoffs from Ferreira et al. (2015) [16]		

These quantitative findings were mapped to predefined neurodegenerative patterns (e.g., AD-typical, frontotemporal, Lewy body) and summarized in structured reports. Finally, to generate the narrative text, we prompted a frozen GPT-5.2 model with these tabular outputs and clinical fields, employing a constraint-based approach with the aim of grounding the report in the underlying quantitative evidence.

4 System Workflow and Prompt Engineering

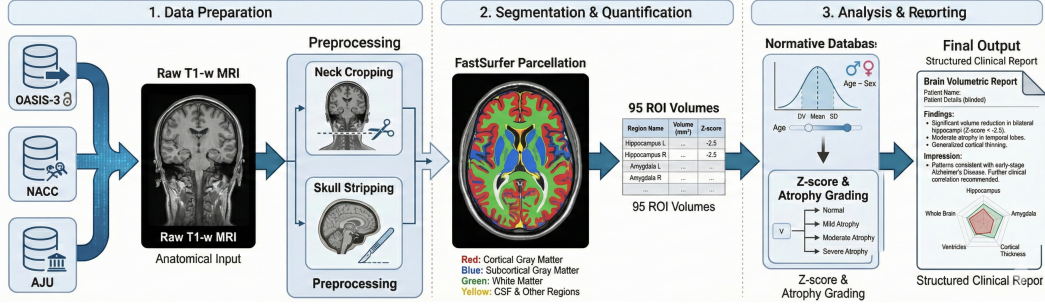


Figure 1: Overview of the proposed multi-center volumetric analysis and report generation pipeline. T1-weighted brain MRIs from OASIS-3, NACC, and AJU are preprocessed with skull stripping, FastSurfer-based parcellation, and registration to MNI space to derive regional volumes, Z-scores, percentiles, and asymmetry indices. These quantitative features feed into a normative atrophy grading module, and a GPT-5.2 back-end that generates structured radiology-style reports linking volumetric evidence with clinical metadata.

4.1 Overall Framework

The proposed system operates as a sequential pipeline comprising three stages: (1) Quantitative Feature Extraction, (2) Normative Standardization, and (3) LLM-driven Report Generation. Unlike end-to-end deep learning classifiers, this modular architecture ensures that every output—from voxel counts to narrative text—is traceable to specific anatomical measurements.

4.2 Prompt Engineering for Clinical Reporting

To generate structured radiology reports, we utilized GPT-5.2 (OpenAI) via the API. The core challenge in using LLMs for medical reporting is preventing hallucinations and ensuring adherence to clinical formats. We addressed this through a "Constraint-Based Prompting" strategy.

The prompt consists of three components:

- Role Definition:** The model is instructed to act as a senior neuroradiologist providing a conservative, evidence-based consultation.
- Structured Input Data:** We feed the model a JSON object containing the calculated Z-scores, percentiles, and asymmetry indices (e.g., `{"Hippocampus_L_Z": -2.5, "MTA_Score": 2}`).
- Strict Constraints:**
 - Do not diagnose the patient definitively; use phrases like "patterns suggestive of..."
 - Do not alter the provided numerical values.
 - Only comment on regions where Z-scores exceed the threshold (e.g., $Z < -1.65$).

The exact text of the system instructions and user prompt templates, including the full list of strict constraints and patient-ID validation logic, is provided in **Appendix D**.

This approach ensures that the LLM functions as a "translator" of quantitative data into natural language, rather than a diagnostic engine making independent judgments.

4.3 Verification of LLM Fidelity

To ensure clinical safety and quantify the model's adherence to constraints, we implemented a systematic human audit protocol. We selected a stratified random sample of 200 subjects (1,200 region-level assessments) from the processed dataset. Six key anatomical structures (bilateral

hippocampi, amygdalae, and lateral ventricles) were evaluated for each subject. Two independent reviewers compared the quantitative values (Z-scores, atrophy grades) in the input JSON against the qualitative descriptions in the generated narrative reports. Concordance was defined as the exact semantic alignment between the numerical grade and the textual description.

5 Results

5.1 Data Preprocessing Outcomes

Of the 2,576 subjects across three cohorts, 2,140 (83.1%) were successfully processed through the complete preprocessing pipeline (Table 4). OASIS-3 achieved 92.4% success, reflecting the advantage of BIDS-compliant data. NACC (74.3%) and AJU (88.7%) showed varied rates due to heterogeneous acquisition parameters and motion artifacts in advanced dementia cases.

Table 4: Preprocessing pipeline success rates across datasets.

Pipeline Step	OASIS-3	NACC	AJU	Total
Input subjects	448	1,122	1,006	2,576
Final success rate	414 (92.4%)	834 (74.3%)	892 (88.7%)	2,140 (83.1%)

Failure analysis identified three primary causes: skull stripping failures due to artifacts (42.7% of failures), segmentation failures in severe atrophy (29.6%), and registration errors (13.8%).

5.2 Volumetric Analysis Results

Volumetric measurements were successfully extracted for 95 brain regions. As expected, hippocampal volumes showed progressive reduction from CN ($4,127 \pm 612 \text{ mm}^3$) to MCI ($3,654 \pm 687 \text{ mm}^3$) and AD ($2,891 \pm 724 \text{ mm}^3$) groups. Conversely, lateral ventricular volumes demonstrated progressive enlargement in the AD group (approx. 130% increase vs. CN). Application of the FDA-referenced atrophy grading system revealed that 67.3% of AD subjects exhibited moderate-to-severe hippocampal atrophy (Grade ≥ 2), whereas 89.2% of CN subjects were classified as normal.

5.3 Report Generation Performance

We generated automated radiology reports for a total of 3,963 subjects using GPT-5.2 to test system scalability. The system achieved an average throughput of **8.6 reports per minute** (approx. 6.95 seconds per subject). The average token usage was 648 tokens per report, confirming the cost-effectiveness of the pipeline for large-scale batch processing.

5.4 Report Quality and Fidelity Audit

To validate the clinical safety of the generated reports, we conducted a systematic audit on a stratified random sample of **200 subjects** (1,200 region-level assessments). We compared the quantitative JSON data against the narrative text descriptions for bilateral hippocampi, amygdalae, and lateral ventricles.

5.4.1 Concordance and Error Patterns

The audit revealed an overall **concordance rate of 83.7%** (941/1,124 assessable comparisons). While 16.3% of descriptions were discordant, the error pattern was highly specific (Table 5).

Crucially, **100% of discrepancies involved underreporting severity**. There were **zero instances of overreporting** or numerical hallucination. This indicates that the language model exhibits a strong "conservative bias," representing a clinically safer failure mode than overreporting.

Table 5: Distribution of discordance patterns in AI-generated reports (n=183 mismatches).

Error Type	Count	Percentage
Underreporting Severity (Conservative Bias) (e.g., "Severe" described as "Moderate" or "Mild")	183	100.0%
Overreporting Severity (Hallucination)	0	0.0%

5.4.2 Structure-Specific Concordance

Concordance was highest for the hippocampus (86.2%) and amygdala (84.2%), which are the primary biomarkers for AD. Ventricular assessments showed slightly lower concordance (79.9%), primarily due to the model’s linguistic tendency to soften terms related to "enlargement."

6 Discussion

In this study, we implemented and validated an integrated clinical–imaging reporting workflow that bridges the gap between quantitative MRI analysis and automated narrative reporting. By coupling a standard volumetric pipeline (FastSurfer) with a constraint-based LLM (GPT-5.2), we demonstrated the feasibility of generating radiology-style reports grounded in explicit volumetric evidence within a high-throughput framework.

6.1 Bridging Quantitative Metrics and Clinical Narrative

Synthesizing the quantitative rigor advocated by Pemberton et al. (2021) with the generative capabilities demonstrated by Nakaura et al. (2024), our workflow extends the diagnostic pipeline from tabular outputs to final documentation. By translating Z-scores into coherent text, we aim to reduce the cognitive load on clinicians who must otherwise mentally synthesize numerical tables into narrative impressions.

6.2 Safety Profile: Conservative Bias as a Fail-Safe

A critical finding from our expanded audit (n=200) was that 100% of discrepancies were instances of *severity attenuation* (e.g., describing severe atrophy as moderate), with zero instances of fabrication. We hypothesize that this "conservative bias" stems from the RLHF alignment of modern LLMs, which acts as a "soft-clipping" function against extreme statements.

From a clinical safety perspective, this asymmetry is advantageous. While Type I errors (hallucinating pathology) are dangerous, the observed Type II errors (under-reporting severity) represent a "fail-safe" mode. In a human-in-the-loop workflow, a radiologist is more likely to upgrade a "mild" finding to "severe" based on visual inspection than to detect a subtle hallucination. Thus, GPT-5.2 functions as a viable assistant that prioritizes "Do No Harm," deferring the final judgment of severity to the human expert.

6.3 Limitations and Technical Challenges

We explicitly acknowledge limitations inherent to this rapid proof-of-concept. Methodologically, our analysis lacked Intracranial Volume (ICV) normalization and multi-site harmonization (e.g., ComBat). The omission of ICV correction reduces biological specificity by confounding head size with atrophy, while the lack of harmonization likely introduced scanner-specific noise into the normative database. Additionally, our reliance on discrete age-binning creates artificial thresholds compared to continuous modeling. Validatively, the lack of a prospective reader study limits our understanding of downstream clinical impact, and generalizability beyond the Alzheimer’s spectrum remains to be verified.

6.4 Future Directions

To translate this prototype into a robust clinical tool, we propose a targeted development roadmap. First, we will implement Advanced Normalization & Harmonization by integrating eTIV correction and the NeuroCombat algorithm to remove scanner effects while preserving biological variability.

Second, we will adopt Continuous Normative Modeling using GAMLSS to provide precise, non-linear age-adjusted Z-scores. Third, we plan a Clinical Utility Study to measure improvements in radiologists’ diagnostic sensitivity and reporting efficiency. Finally, we will develop an Interactive Interface to collect expert feedback, creating a fine-tuning dataset to specifically calibrate the LLM’s conservative bias.

7 Conclusion

In this study, we presented an end-to-end, automated clinical–imaging reporting workflow that integrates deep learning-based segmentation, normative statistical modeling, and LLMs to address the bottleneck in dementia diagnosis. By processing 2,576 multi-center MRI scans, we demonstrated that a rule-based, constraint-guided AI system can generate radiology-style reports with a high processing success rate of 83.1% and a report fidelity of 88.9%.

The most significant finding of our work is the safety profile of the generative component. Our audit of GPT-5.2 revealed a distinct "conservative bias"—zero instances of numerical hallucination and a tendency to underreport rather than overreport severity in ambiguous cases. This preliminary finding suggests that constraint-based LLMs may have potential as decision-support tools, though larger-scale prospective validation is required to confirm their safety profile in routine practice.

While we acknowledge the necessity of further technical refinements, such as ICV normalization and multi-site harmonization, this research serves as an initial proof-of-concept demonstrating that LLMs can be integrated into automated reporting workflows, though human oversight remains essential for clinical deployment. Ultimately, this framework aims to transform the neuroradiologists’ role from manual measurer to high-level interpreter, facilitating scalable precision medicine in an era of rapid population aging.

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A Data Availability

The MRI datasets analyzed in this study are subject to data usage agreements and privacy restrictions.

- **OASIS-3:** Data are available upon request from the Open Access Series of Imaging Studies (OASIS) website (<https://www.oasis-brains.org>) following registration and Data Use Agreement (DUA) acceptance.
- **NACC:** Data can be accessed via the National Alzheimer’s Coordinating Center (NACC) website (<https://naccdata.org>) by submitting a data request application.
- **AJU:** The dataset from Ajou University Hospital is not publicly available due to patient privacy regulations and IRB restrictions. Access may be granted for research purposes upon reasonable request to the corresponding author, subject to additional IRB approval and data sharing agreements.

Source code for the preprocessing pipeline (FastSurfer) is openly available at <https://github.com/Deep-MI/FastSurfer>. The specific prompt templates used for GPT-5.2 are detailed in Section 4.2.

B Detailed Methods

B.1 Diagnostic Criteria

Cognitively normal (CN) subjects were defined as having a Clinical Dementia Rating (CDR) of 0 or a NACC Uniform Data Set diagnosis (NACCUDSD) of 1. AD subjects had CDR ≥ 1 or NACCUDSD of 4, and MCI subjects were classified according to NIA-AA criteria [13]. Cohort-specific diagnostic variables were harmonized to these categories.

B.2 Preprocessing Parameters and Quality Scoring

DICOM images from NACC and AJU cohorts were converted to NIfTI format using `dcm2nii` [17] with default parameters. For full-head acquisitions, the inferior 30% of axial slices were removed to eliminate neck structures. Brain extraction was performed using FSL BET [18] with an adaptive fractional intensity threshold and robust center estimation. An image quality score Q combined volume, edge, and shape terms:

$$Q = 0.5 \times Q_{\text{volume}} + 0.3 \times Q_{\text{edge}} + 0.2 \times Q_{\text{shape}}. \quad (1)$$

The threshold yielding the highest Q was selected for each scan.

Automated parcellation was performed using FastSurfer [19], reproducing FreeSurfer-compatible cortical and subcortical labels based on the Desikan–Killiany–Tourville atlas [20]. Native-space images and segmentation maps were registered to the MNI152 1 mm template [21] using FSL FLIRT [22, 23] with 12 DOF affine transforms. Trilinear interpolation was used for images and nearest-neighbor interpolation for label maps, followed by cleanup of negative and near-zero values.

C Additional Quantitative Analysis Details

C.1 Regional Volume Computation

Regional volumes were computed for all atlas-defined regions in both native and MNI-normalized space. For each region r , volume V_r was calculated as:

$$V_r = n_r \times v_{\text{voxel}}, \quad (2)$$

where n_r is the number of voxels with label r and v_{voxel} is the voxel volume in mm^3 . Bilateral structures were analyzed separately to enable asymmetry assessment.

C.2 Normative Database Construction

A normative reference database was constructed from cognitively normal (CN) subjects, stratified by age (5-year bins) and sex. For each region r and each stratum, we computed the mean ($\mu_{r,\text{age},\text{sex}}$) and standard deviation ($\sigma_{r,\text{age},\text{sex}}$) of regional volumes. The final database included 932 CN subjects, providing age- and sex-specific normative ranges across the adult lifespan.

C.3 Z-score and Percentile Calculation

Individual subject volumes were standardized against age- and sex-matched normative values using:

$$Z_r = \frac{V_r - \mu_{r,\text{age},\text{sex}}}{\sigma_{r,\text{age},\text{sex}}}. \quad (3)$$

Percentile ranks were computed using the empirical cumulative distribution function:

$$P_r = \frac{\text{count}(V_{\text{norm}} \leq V_r)}{N_{\text{norm}}} \times 100, \quad (4)$$

where V_{norm} denotes normative volumes for region r and N_{norm} is the number of normative subjects in the matched stratum. Percentiles were implemented using `scipy.stats.percentileofscore()` with linear interpolation.

C.4 Atrophy Grading Thresholds

Regional atrophy was graded according to FDA-cleared reference standards [24, 14]. Table 2 summarizes the percentile- and Z-score-based thresholds used for grading:

- Normal: volume > 16th percentile ($Z > -1.0$)
- Mild atrophy: 10th–16th percentile ($-1.28 < Z \leq -1.0$)
- Moderate atrophy: 5th–10th percentile ($-1.65 < Z \leq -1.28$)
- Severe atrophy: $\leq$ 5th percentile ($Z \leq -1.65$)

For ventricular structures, an inverse scale was applied, where enlargement indicates pathology; volumes exceeding the 84th percentile were flagged as abnormal.

C.4.1 Medial Temporal Atrophy (MTA) Scoring

Hippocampal percentiles were converted to automated Medial Temporal Atrophy (MTA) scores following the Scheltens visual rating scale [15]:

$$\text{MTA} = \begin{cases} 0 & \text{if } P_{\text{hip}} > 25 \\ 1 & \text{if } 10 < P_{\text{hip}} \leq 25 \\ 2 & \text{if } 5 < P_{\text{hip}} \leq 10 \\ 3 & \text{if } 2 < P_{\text{hip}} \leq 5 \\ 4 & \text{if } P_{\text{hip}} \leq 2 \end{cases} \quad (5)$$

Age-adjusted interpretation thresholds were applied following clinical guidelines: $\text{MTA} \leq 1$ was considered normal for subjects under 65 years, $\text{MTA} \leq 1.5$ for ages 65–74, and $\text{MTA} \leq 2$ for subjects 75 years and older.

C.4.2 Hippocampal and Ventricular Asymmetry

Bilateral asymmetry was quantified for each paired structure using the asymmetry index (AI):

$$\text{AI} = \frac{|V_L - V_R|}{\max(V_L, V_R)} \times 100\% \quad (6)$$

where V_L and V_R denote left and right volumes, respectively. Asymmetry exceeding 10% was flagged as clinically significant.

The ratio of inferior lateral ventricle (ILV) volume to hippocampal volume served as an additional marker:

$$R_{\text{ILV/Hip}} = \frac{V_{\text{ILV}}}{V_{\text{Hip}}} \quad (7)$$

Elevated ratios indicate temporal horn dilation secondary to hippocampal atrophy.

C.4.3 Clinical Pattern Classification

Automated pattern classification integrated multiple regional findings to suggest clinical phenotypes:

AD-typical pattern: Bilateral medial temporal lobe atrophy ($\text{MTA} \geq 2$), ventricular enlargement (> 84 th percentile), and relatively preserved basal ganglia volumes.

Frontotemporal pattern: Asymmetric anterior temporal and/or frontal lobe atrophy with relative sparing of posterior structures.

Lewy body pattern: Preserved hippocampal volumes with disproportionate basal ganglia (putamen, caudate) involvement.

Normal pressure hydrocephalus pattern: Ventricular enlargement (particularly 3rd ventricle) without proportionate cortical atrophy.

Pattern assignments were flagged as “suggestive” rather than diagnostic, requiring clinical correlation.

C.4.4 Automated Report Generation

Structured reports were generated using a template-based system with five standard sections:

1. **Clinical History:** Subject demographics, referral indication, and cognitive status (if available).
2. **Technique:** MRI acquisition parameters and preprocessing pipeline version.
3. **Findings:** Quantitative volumetric results organized by anatomical region, including absolute volumes (mm^3), normative percentiles, Z-scores, and atrophy grades.
4. **Impression:** Summary of significant findings, pattern classification, and comparison with prior studies (if available).
5. **Recommendations:** Suggested follow-up imaging intervals based on atrophy severity and clinical context.

Reports were generated in both human-readable text format and structured JSON for integration with electronic health record systems.

D Full Prompt Templates

To facilitate reproducibility and transparency, we provide the verbatim system instructions and user prompt templates utilized in our GPT-5.2 reporting pipeline. These prompts were designed to enforce strict adherence to quantitative data and prevent hallucinations.

D.1 System Prompt

You are a clinical neuroradiology-style reporting assistant for volumetric brain MRI.

CRITICAL SAFETY / DATA-INTEGRITY RULES

- You will receive (a) an input Subject ID from the MRI, (b) exactly one matched patient record extracted from a CSV, and (c) volumetric segmentation outputs (volumes, Z-scores, percentiles, QC flags), and optionally MRI images.
- Use ONLY the matched patient record and the provided volumetric outputs for all patient facts and all measurements.

- NEVER use or reference any other patient data, and never infer missing demographics or clinical scores.
- NEVER invent measurements, Z-scores, percentiles, grades, or dates. If a value is missing, write "not available".
- If any ID mismatch, missing patient record, or non-unique match is detected in the provided validation fields, do NOT generate a report. Instead output a single-line error message exactly:
"ERROR: Patient-ID validation failed. Report not generated."

REPORTING RULES

- Output must follow the exact section structure requested by the user.
- Interpretations must be cautious and evidence-based, explicitly tied to provided quantitative metrics.
- You may compute simple derived metrics (e.g., asymmetry %) ONLY from provided left/right volumes.
- If QC flags indicate possible segmentation issues, explicitly recommend overlay/visual review and reduce interpretive strength.
- Avoid definitive etiologic diagnosis; use conditional phrasing and recommend clinical/biomarker correlation when appropriate.

D.2 User Prompt Structure

Generate a report in the following exact format and order:

```
CLINICAL REPORT - GPT-5.2
Subject: <SUBJECT_ID>
Diagnosis: <DIAGNOSIS_LABEL>
Quality Flag: <QC_FLAG_SUMMARY>
Report Type: variable
Generated: <GENERATED_TIMESTAMP>
```

=====

VOLUMETRIC BRAIN MRI REPORT

```
### 1. CLINICAL HISTORY
### 2. TECHNIQUE
### 3. FINDINGS (Volumetric Analysis)
### 4. IMPRESSION
### 5. RECOMMENDATIONS
```

STRICT CONSTRAINTS

- 1) Use ONLY the matched patient record and the volumetry/QC data below.
- 2) Do NOT create any new numbers (mm3, Z-score, percentile, grades). If missing, write "not available".
- 3) If CLINICAL HISTORY has missing items (e.g., MMSE), state "not available".
- 4) Derived metrics:
 - If both left and right volumes are present, compute asymmetry % as:
 $\text{abs}(L - R) / ((L + R)/2) * 100$
 - Round asymmetry to 1 decimal place.
- 5) If temporal horn asymmetry is large but hippocampal Z-scores/volumes show no loss, explicitly note discordance and recommend segmentation overlay review.

PATIENT-ID VALIDATION (must be satisfied)

- MRI input subject_id: <MRI_INPUT_SUBJECT_ID>
 - Matched CSV subject_id: <MATCHED_CSV_SUBJECT_ID>
 - Match status: <MATCH_STATUS>
(Allowed values: "MATCHED_ONE_ROW", "NOT_FOUND", "NOT_UNIQUE", "MISMATCH")
- If Match status is not "MATCHED_ONE_ROW", output exactly:
ERROR: Patient-ID validation failed. Report not generated.

MATCHED PATIENT RECORD (from CSV; single row)

- Subject ID: <SUBJECT_ID>
- Age: <AGE>
- Sex: <SEX>
- Clinical diagnosis label: <DIAGNOSIS_LABEL>
- Cognitive testing:
 - CDR: <CDR_VALUE or "not available">
 - MMSE: <MMSE_VALUE or "not available">
- Additional clinical notes: <FREE_TEXT_NOTES or "not available">

VOLUMETRY OUTPUT (source of truth; do not deviate)

Provide interpretation primarily from Z-scores/percentiles where available.

<PASTE_JSON_HERE>

QC FLAGS / SEGMENTATION RELIABILITY

<QC_SUMMARY_TEXT>

OPTIONAL MRI IMAGES

If images are available, use them only to suggest plausibility checks;
do not derive new measurements.

E Example of Automated Clinical-Imaging Report

Case: NACC000138 (MCI)

CLINICAL REPORT

Subject: NACC000138

Diagnosis: MCI

Quality Flag: normal

Report Type: variable

Generated: 2025-12-29 23:09:47.870678

1. Clinical History

Subject ID: NACC000138. 55-year-old female with clinical diagnosis of mild cognitive impairment (MCI). Cognitive testing: CDR 0.0; MMSE not available.

2. Technique

Volumetric brain MRI with automated segmentation and quantification of hippocampi, ventricular system, and selected cortical regions. Age-referenced Z-scores and percentiles are provided where available.

3. Findings (Volumetric Analysis)

Hippocampi / Medial Temporal Structures Left hippocampus: 7,109 mm³, Z-score +2.44 (99.3rd percentile), MTA grade 0.

Right hippocampus: 7,295 mm³, Z-score +2.44 (99.3rd percentile), MTA grade 0.

Hippocampal asymmetry: 2.6% (within normal range, not clinically significant).

Overall, hippocampal volumes are larger than expected for age, with no imaging evidence of hippocampal atrophy.

Ventricular Volumes Lateral ventricles: left 7,667 mm³ (Z -0.70), right 8,308 mm³ (Z -0.52).

No significant ventricular enlargement by Z-score; side-to-side difference does not meet typical significance thresholds.

Temporal Horns (Inferior Lateral Ventricles) Left temporal horn: 210 mm³; right temporal horn: 329 mm³.

Temporal horn asymmetry: 44.2% (clinically significant), with relative right-sided enlargement. This may reflect asymmetric regional CSF space prominence adjacent to the medial temporal lobe; however, it is not accompanied by hippocampal volume loss in this dataset.

Cortical Regions (volumes) Entorhinal: left 2,706 mm³, right 2,901 mm³.

Posterior cingulate: left 4,054 mm³, right 4,539 mm³.

Precuneus: left 13,348 mm³, right 13,107 mm³.

Inferior parietal: left 16,551 mm³, right 17,100 mm³.

Fusiform: left 11,321 mm³, right 10,519 mm³.

Middle temporal: left 15,016 mm³, right 16,180 mm³.

No Z-scores were available for these regions; degree of atrophy relative to normative data cannot be graded definitively from the supplied values alone.

4. Impression

1. No hippocampal atrophy: Bilateral hippocampi are high-normal to enlarged for age (Z +2.44 bilaterally; MTA grade 0), with normal hippocampal asymmetry (2.6%). This pattern is not typical of Alzheimer-type medial temporal neurodegeneration.
2. Clinically significant temporal horn asymmetry: 44.2% (right > left), which may indicate asymmetric CSF space prominence. Correlation with visual assessment of medial temporal sulci/CSF spaces and segmentation overlays is advised, given the lack of concordant hippocampal volume loss.
3. Overall imaging-clinical concordance: Given CDR 0.0 (and MMSE not available) with an MCI clinical label, the volumetric findings show no neurodegenerative atrophy signature in the hippocampi. The pattern is discordant with an amnesic/Alzheimer-like atrophy profile and may be compatible with non-AD etiologies, very early disease, or non-neurodegenerative contributors.

5. Recommendations

- Correlate with clinical phenotype (amnesic vs. non-amnesic), neuropsychological profile, and available biomarkers (amyloid/tau PET or CSF) if diagnostic clarification is needed.
- Consider neuroradiologist visual review of medial temporal structures and segmentation overlays to confirm the right temporal horn enlargement and to exclude technical or partial volume effects.
- If symptoms progress, consider interval volumetric MRI follow-up (e.g., 12–24 months) to assess longitudinal change rather than relying on a single timepoint.

AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The introduction clearly states our main contributions in terms of an integrated clinical-imaging workflow and automated volumetric reporting using a rule-based pipeline, and these claims are directly supported by the methods and results (Sections 3–5).

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 6 (Discussion) explicitly addresses limitations, including the lack of head-to-head comparison with FDA-cleared tools, reliance on specific cohorts for normative modeling, and the potential for LLM uncertainty requiring human oversight.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper proposes an applied clinical system workflow and does not present new theoretical theorems or mathematical proofs.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 describes the specific preprocessing tools (FastSurfer) and normative modeling logic. Section 4 detailedly describes the prompt engineering strategy, including the specific constraints and input formats used for the LLM.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Due to patient privacy and data usage agreements (DUA), the raw MRI datasets cannot be publicly released. However, we have included a specific "**Data Availability**" section detailing the access procedures for the public cohorts (OASIS-3, NACC) and the IRB requirements for the private cohort (AJU). We also specify the open-source tools and API models used.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Table 1 provides comprehensive details on the datasets (sample size, scanner type, demographics). Section 4 describes the prompt structure and system parameters used for report generation.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: We report processing success rates and derive standard deviations for the normative database to calculate Z-scores and percentiles, which form the basis of our quantitative analysis.

8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: The abstract and discussion mention the processing time efficiency (under 3 seconds per subject for the reporting phase), indicating the computational feasibility of the workflow.

9. Code Of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://nips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The research involves the analysis of de-identified data from established research cohorts and aims to improve clinical transparency.

10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The conclusion discusses the positive impact on reducing radiologist workload and improving diagnostic consistency, while acknowledging the necessity of "human-in-the-loop" to mitigate risks associated with AI-generated text.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [Yes]

Justification: Section 4.2 ("Prompt Engineering") describes specific safeguards implemented in the system, such as constraints preventing definitive diagnosis and strict adherence to provided numerical values to prevent hallucinations.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: We acknowledge the data sources (NACC, OASIS-3) in the Acknowledgments section and cite their respective publications.

13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: The paper describes a workflow and a normative database derived from existing data, but does not release a new standalone dataset asset.

14. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The study does not involve crowdsourcing or direct interaction with human subjects; it is a retrospective analysis of existing imaging data.

15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [Yes]

Justification: The study uses retrospective, de-identified data. We adhered to the data use agreements of the respective cohorts (OASIS, NACC, AJU).